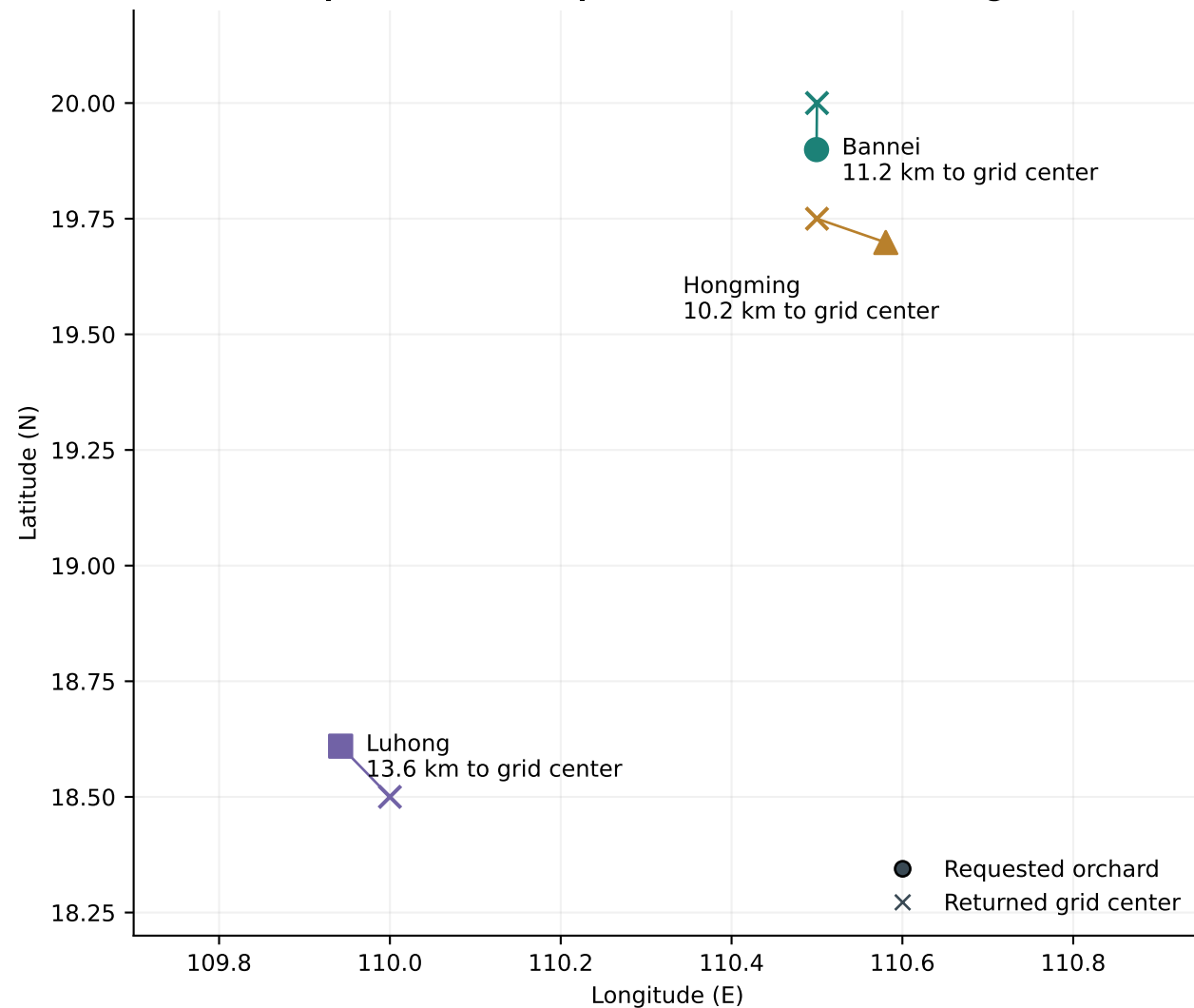


# R4 evidence chain: precise requests do not mean on-site observations

## Requested orchard points and returned ERA5 grids



R4 daily + hourly reanalysis  
3 grids; 5 harvest years



P1: flush to panicle  
Weather does not beat history



P2: panicle to full bloom  
GDD has preliminary held-out support



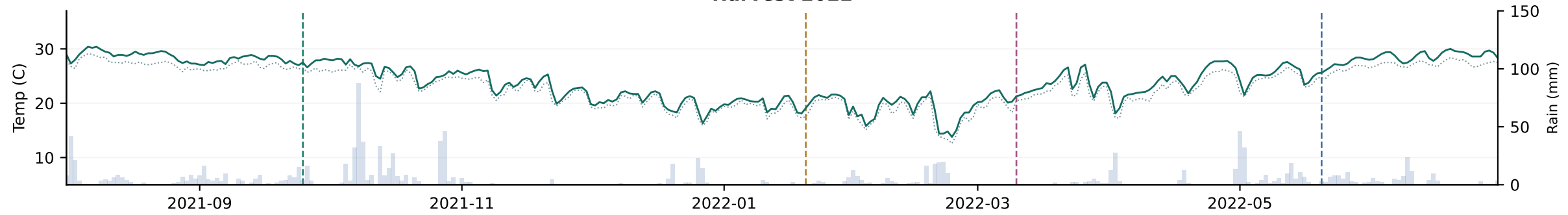
P3: full bloom to maturity  
Small, fragile GDD improvement



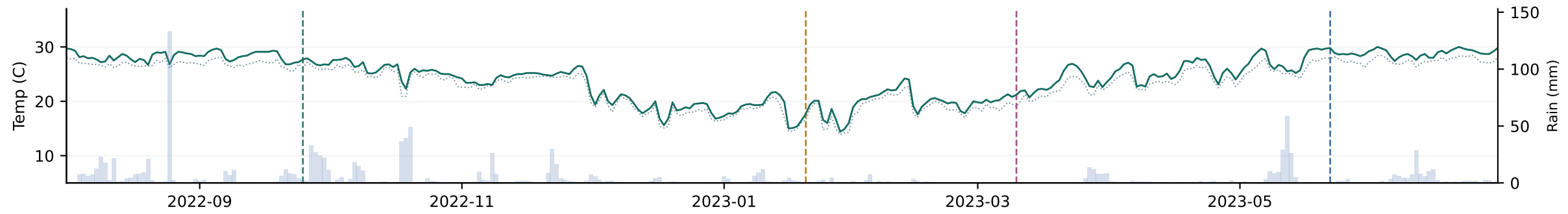
Four weather-window definitions  
Normal yield: exploratory evidence

# Bannei: observed phenology with complete R4 weather, 2022-2026

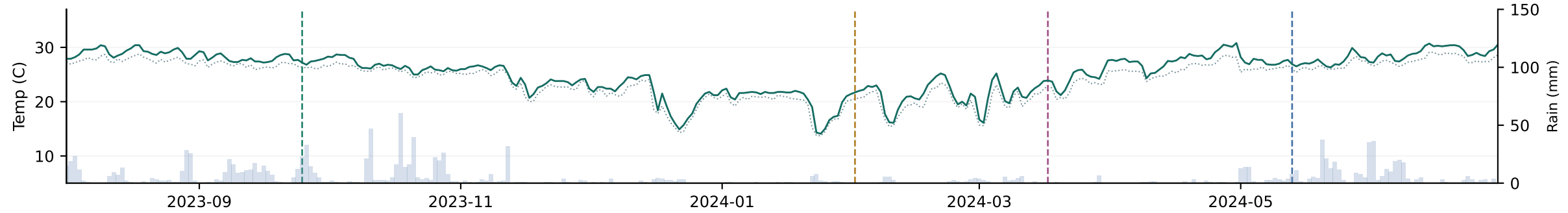
## Harvest 2022



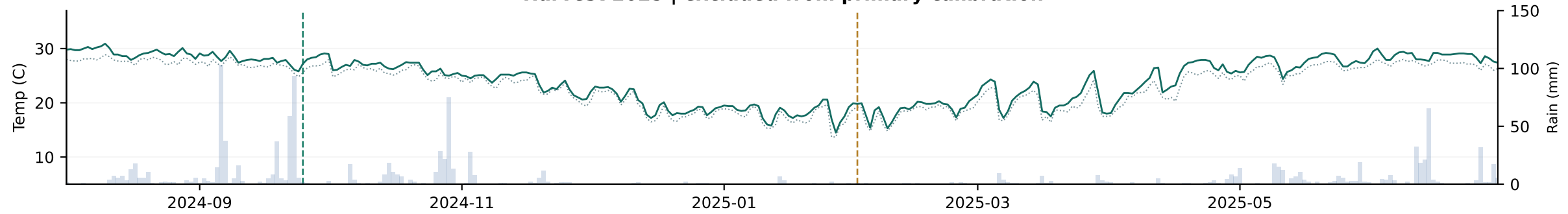
## Harvest 2023



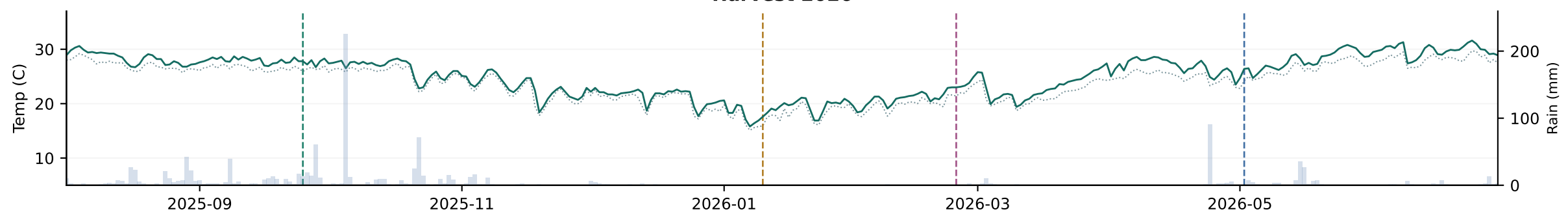
## Harvest 2024



## Harvest 2025 | excluded from primary calibration



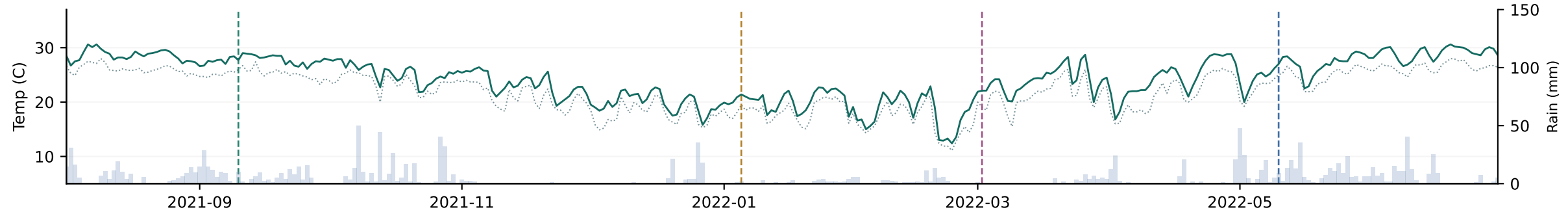
## Harvest 2026



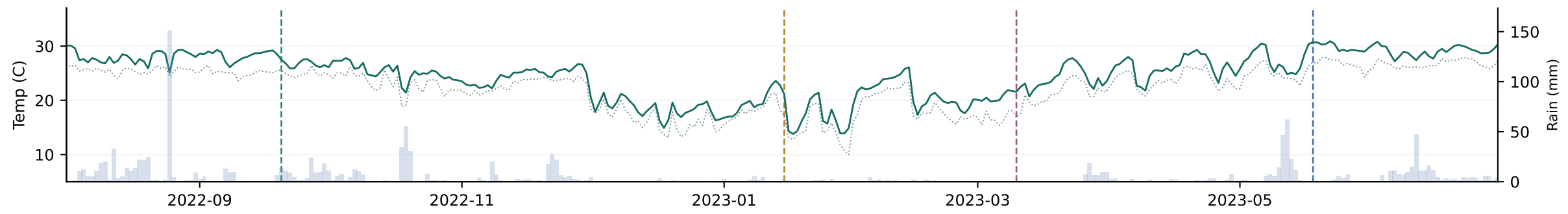
— Daily mean T    ···· Daily minimum T    Rainfall    — Flush    — Panicle    — Full bloom    — Maturity

# Hongming: observed phenology with complete R4 weather, 2022-2026

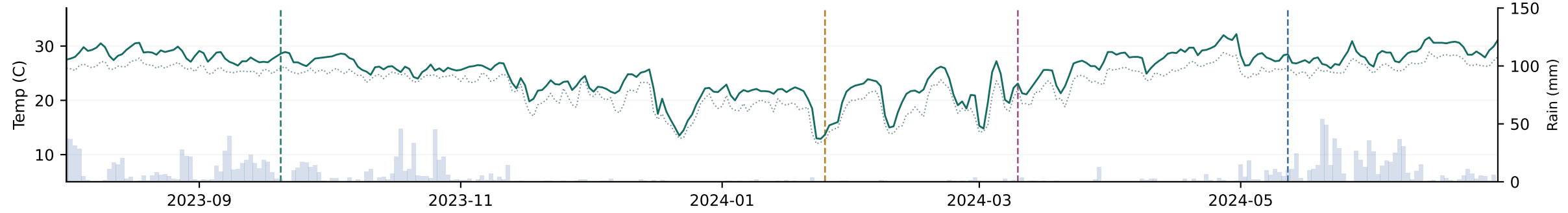
## Harvest 2022



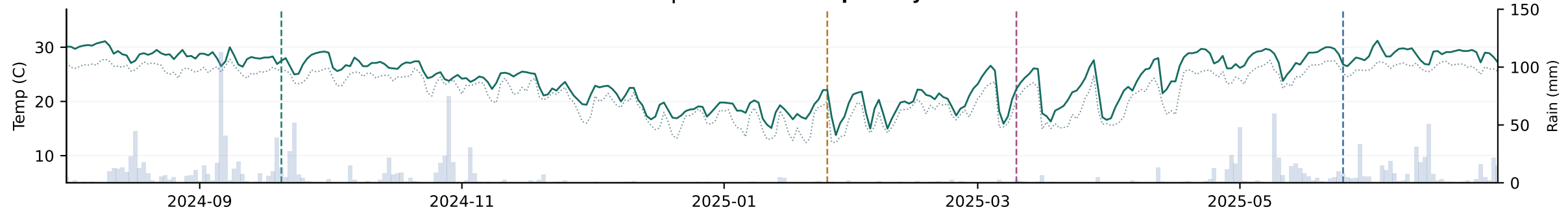
## Harvest 2023



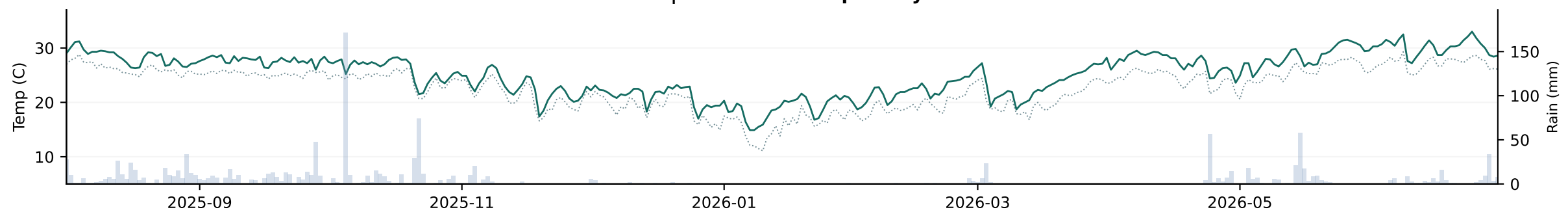
## Harvest 2024



## Harvest 2025 | excluded from primary calibration



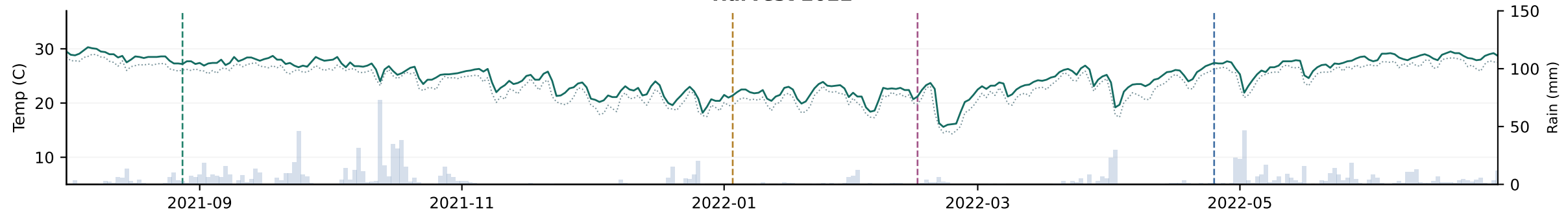
## Harvest 2026 | excluded from primary calibration



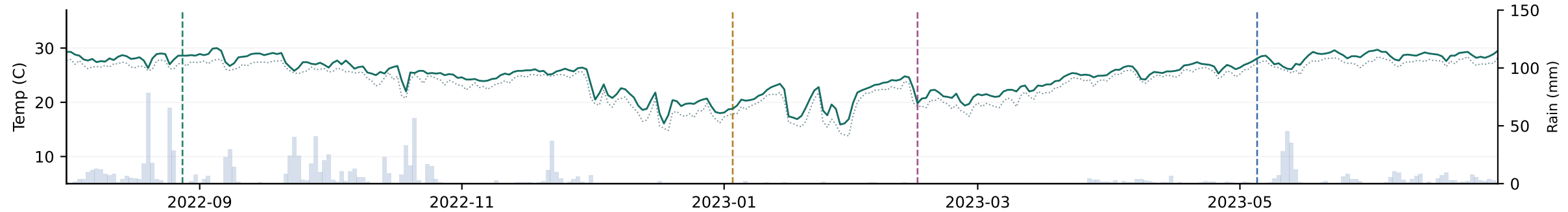
— Daily mean T    ···· Daily minimum T    Rainfall    - - - Flush    - - - Panicle    - - - Full bloom    - - - Maturity

# Luhong: observed phenology with complete R4 weather, 2022-2026

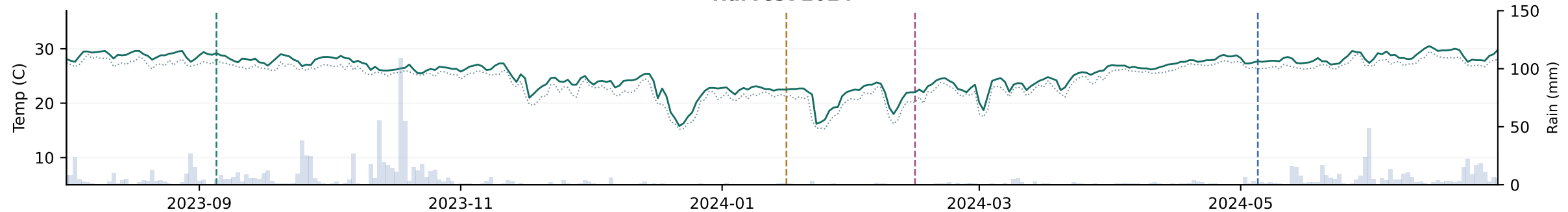
## Harvest 2022



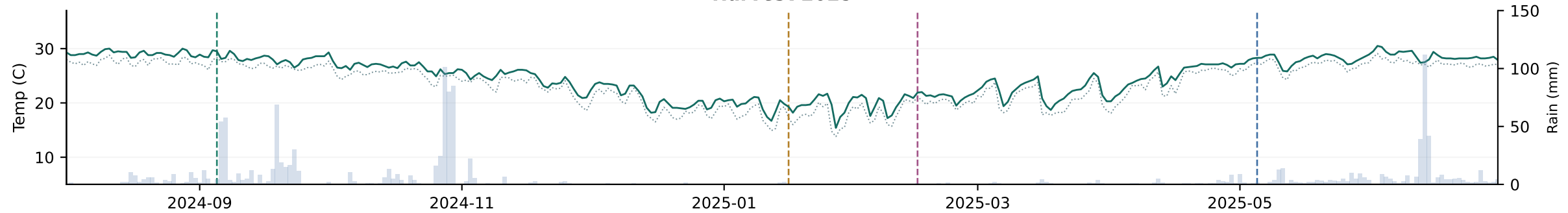
## Harvest 2023



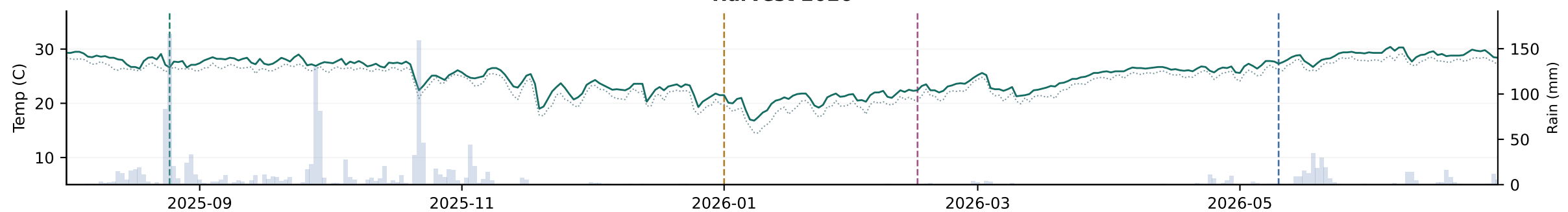
## Harvest 2024



## Harvest 2025

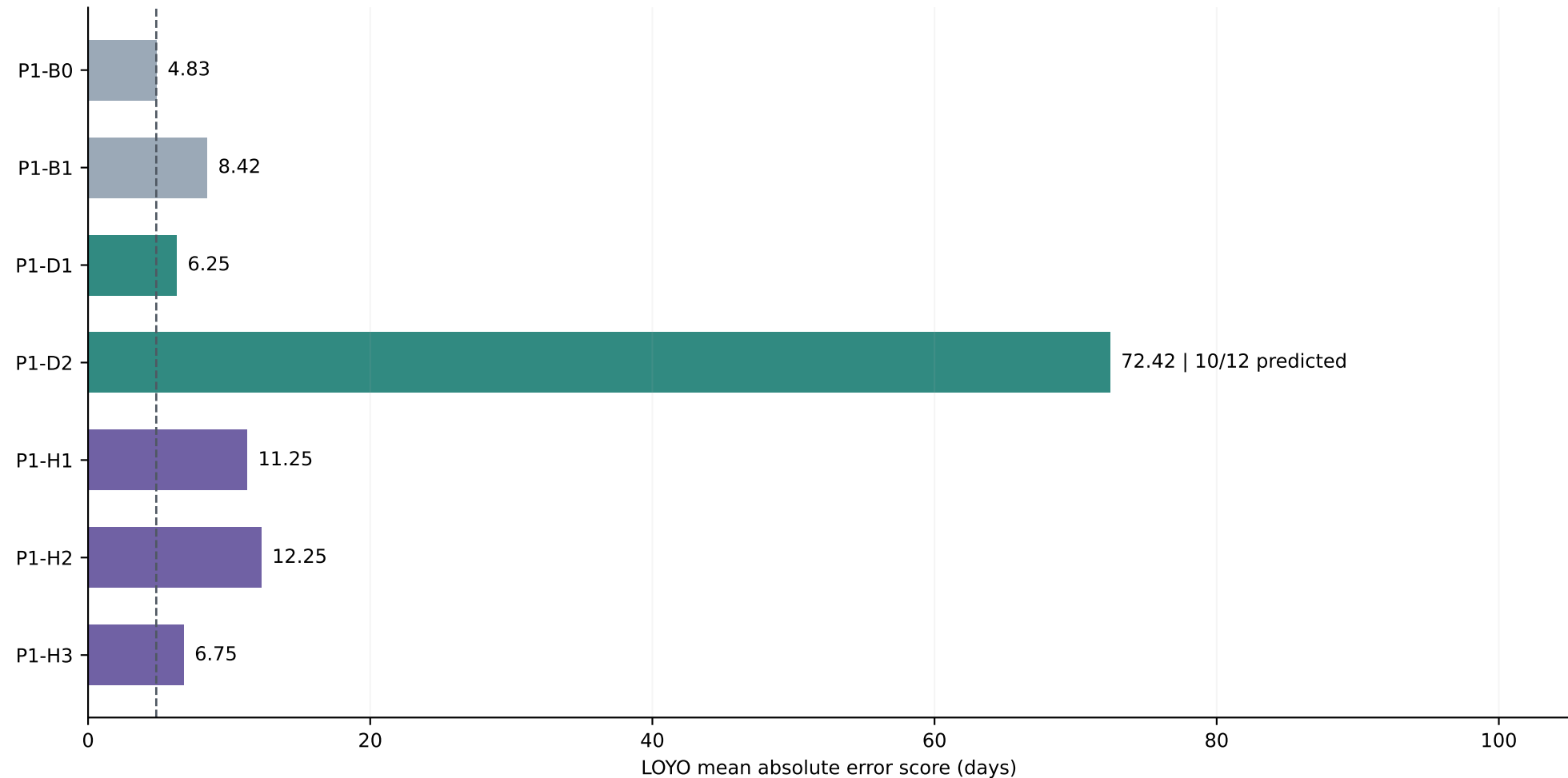


## Harvest 2026



— Daily mean T    ···· Daily minimum T    Rainfall    — Flush    — Panicle    — Full bloom    — Maturity

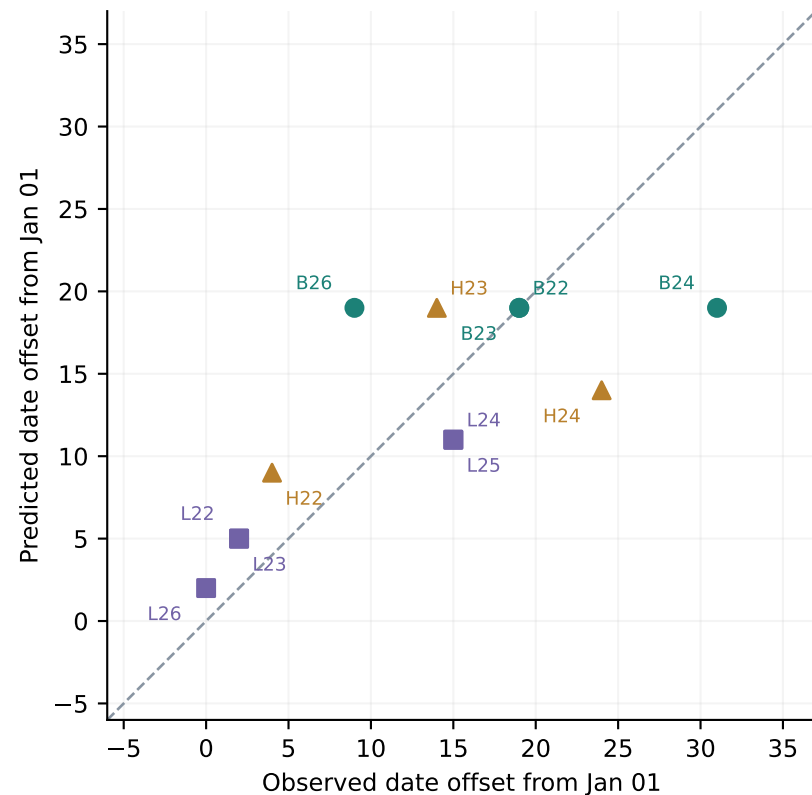
## P1: daily and hourly weather do not outperform the duration baseline



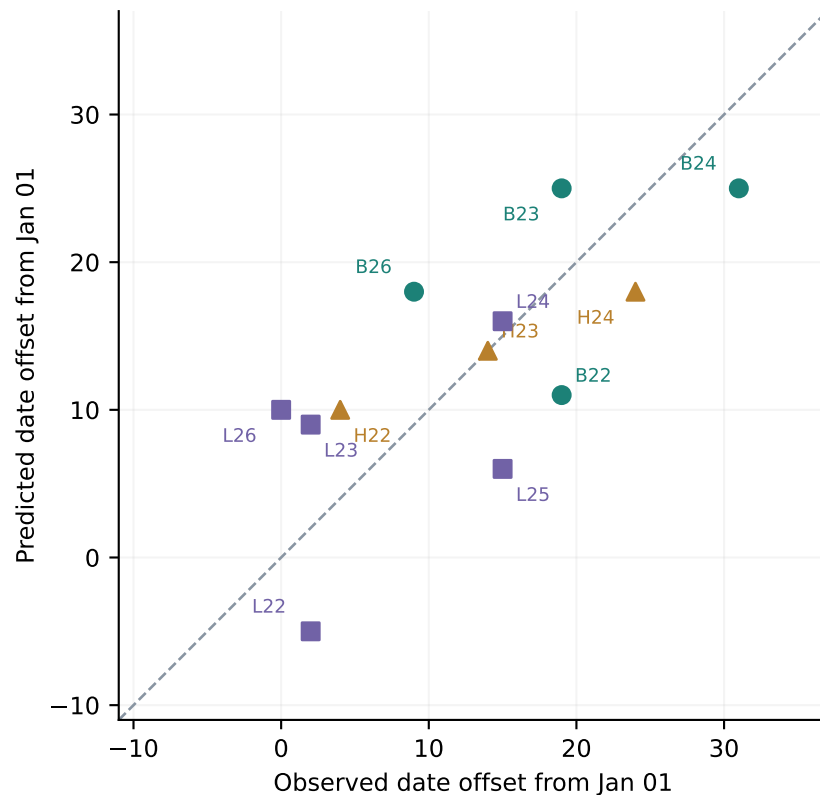
All models evaluated on the same 12 seasons. D2: two non-crossings receive the prespecified 365-day penalty.

P1 held-out dates: B= Bannei, H= Hongming, L= Luhong; suffix = harvest year

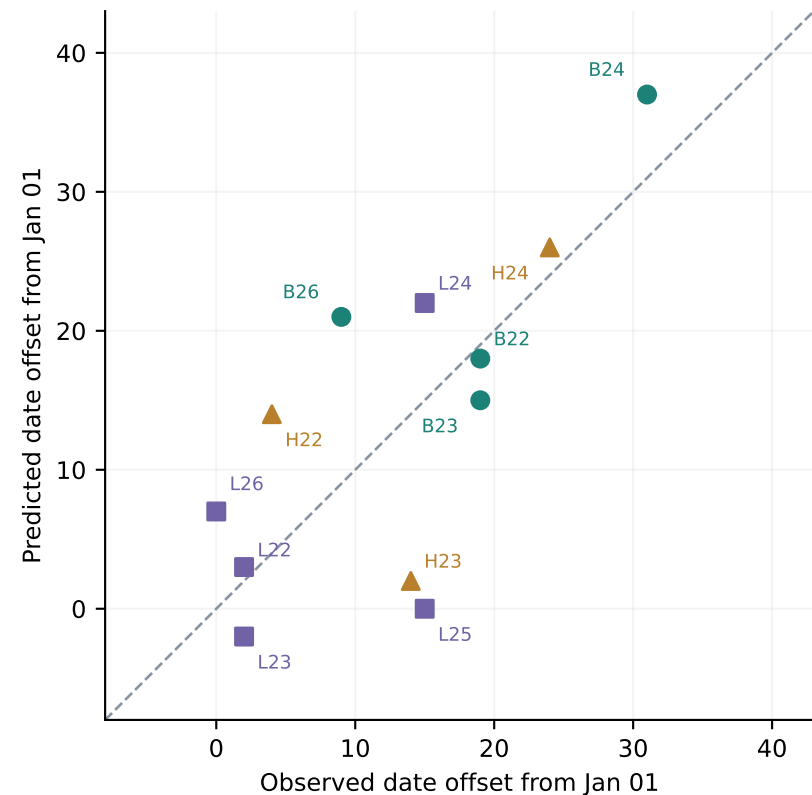
P1-B0



P1-D1

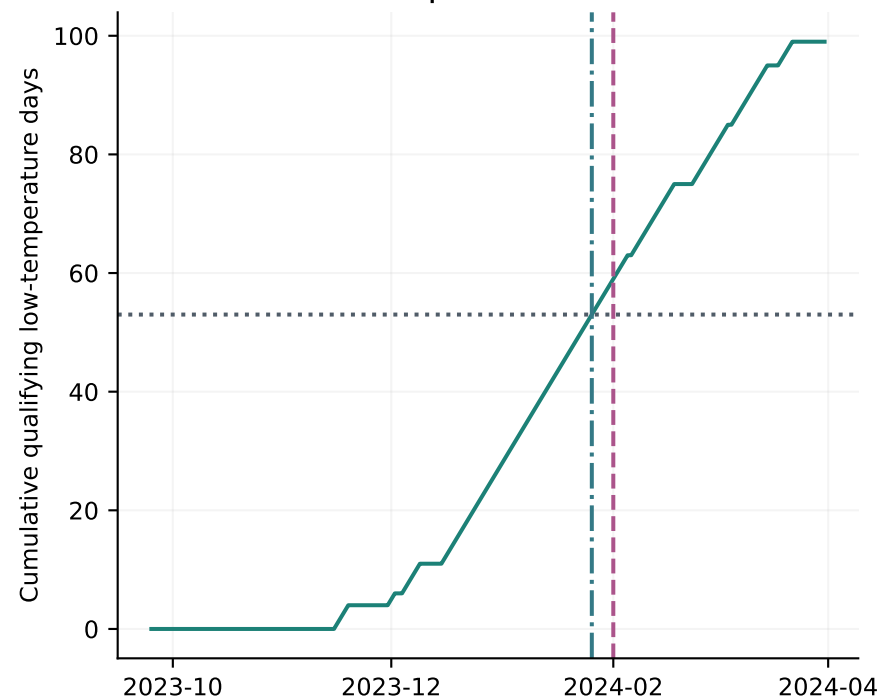


P1-H3

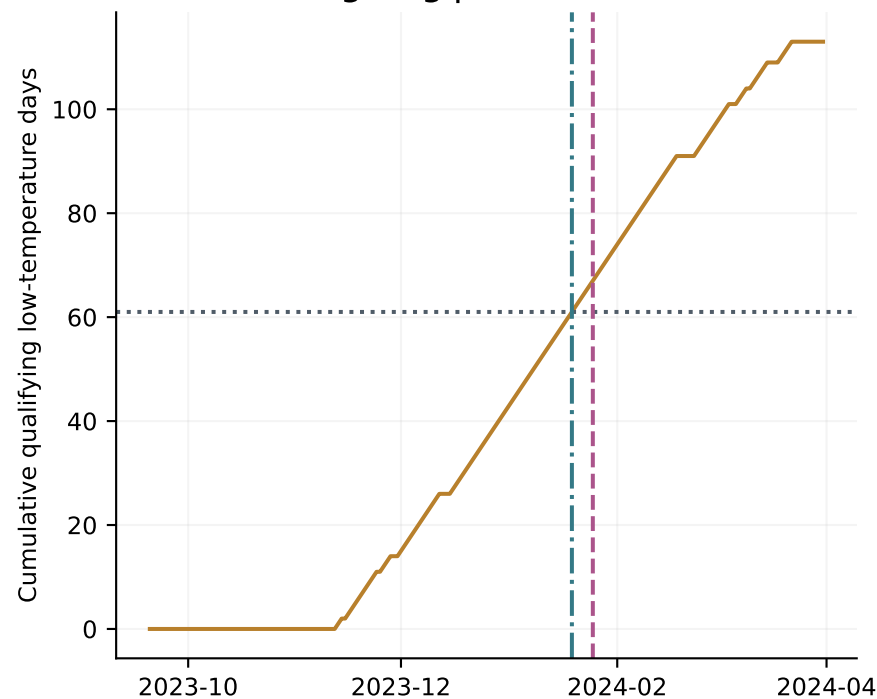


# Illustrative 2024 holdout: P1-D1 accumulations, not a validated physiological mechanism

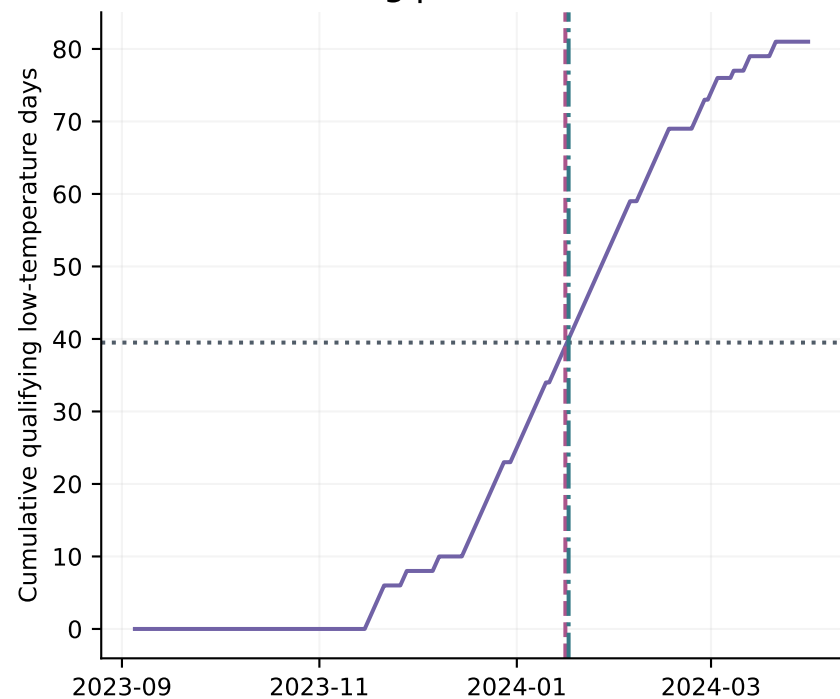
## Bannei | held out 2024



## Hongming | held out 2024



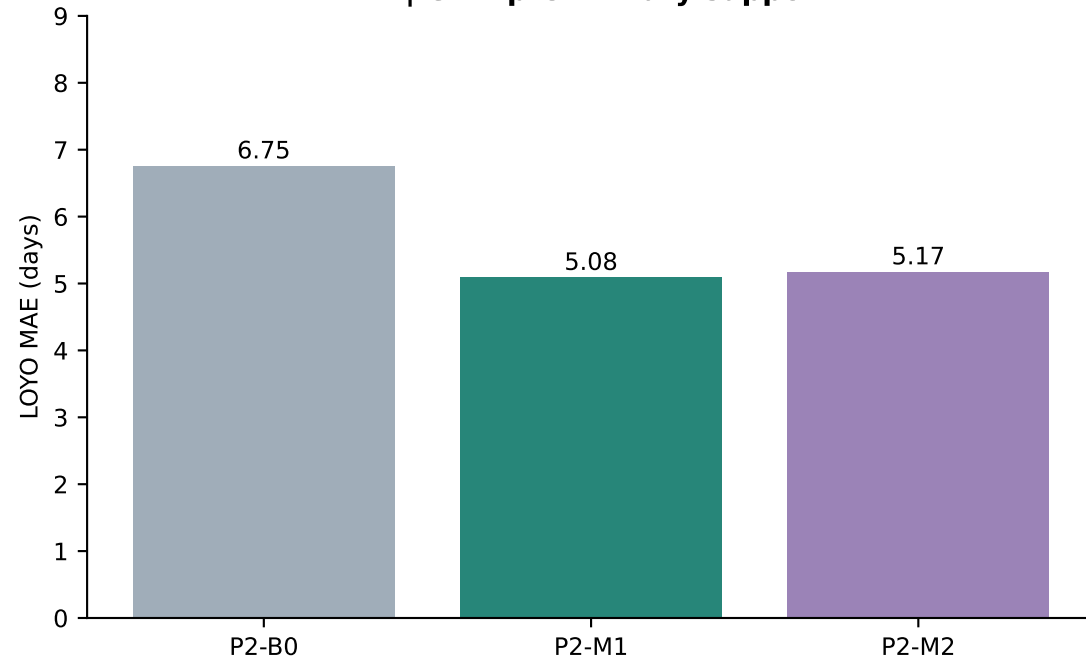
## Luhong | held out 2024



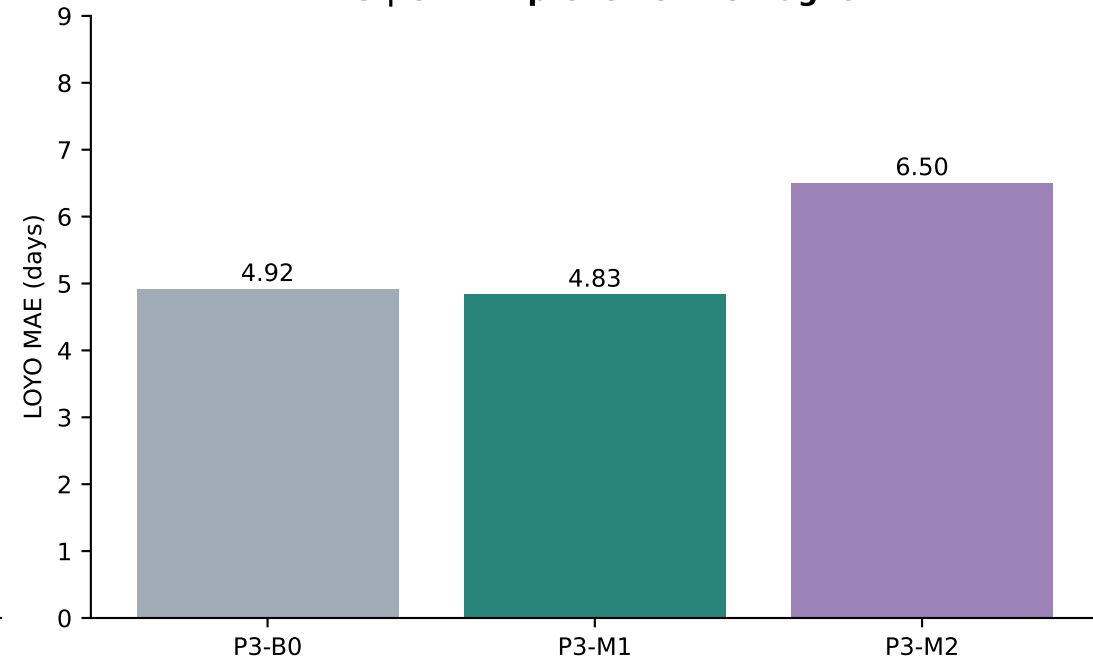
..... Training-only threshold    - - - Observed panicle    - . - Held-out predicted panicle

# Later phenology: stage-specific evidence, not proof of an end-to-end forecast

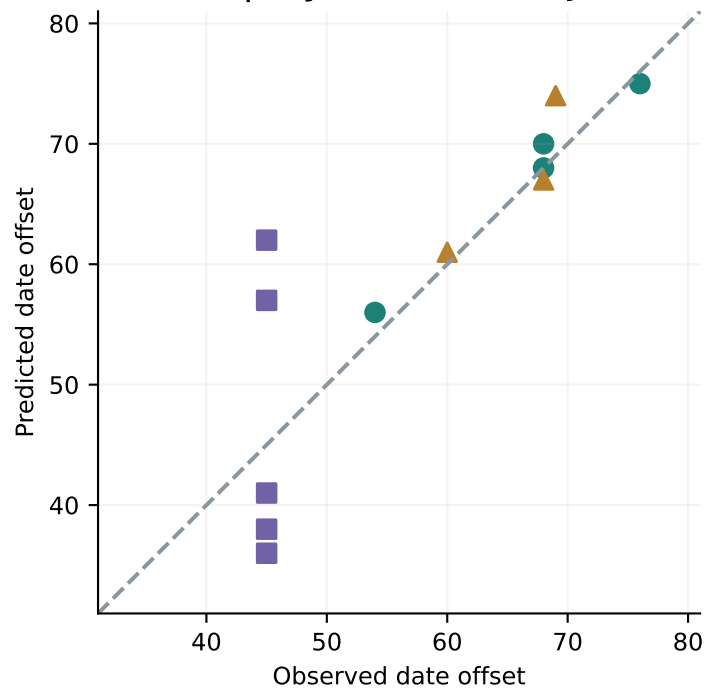
## P2 | GDD preliminary support



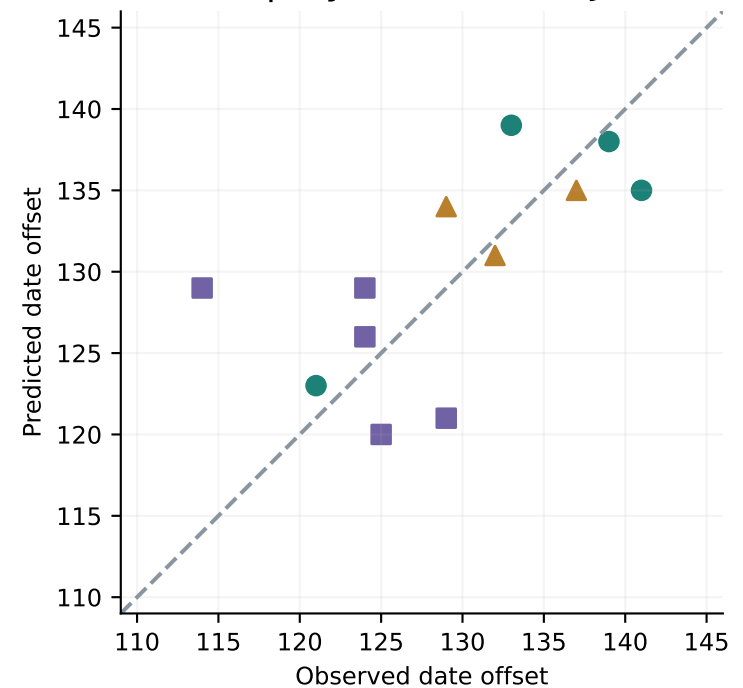
## P3 | GDD improvement is fragile



## P2-M1 | days from harvest Jan 01

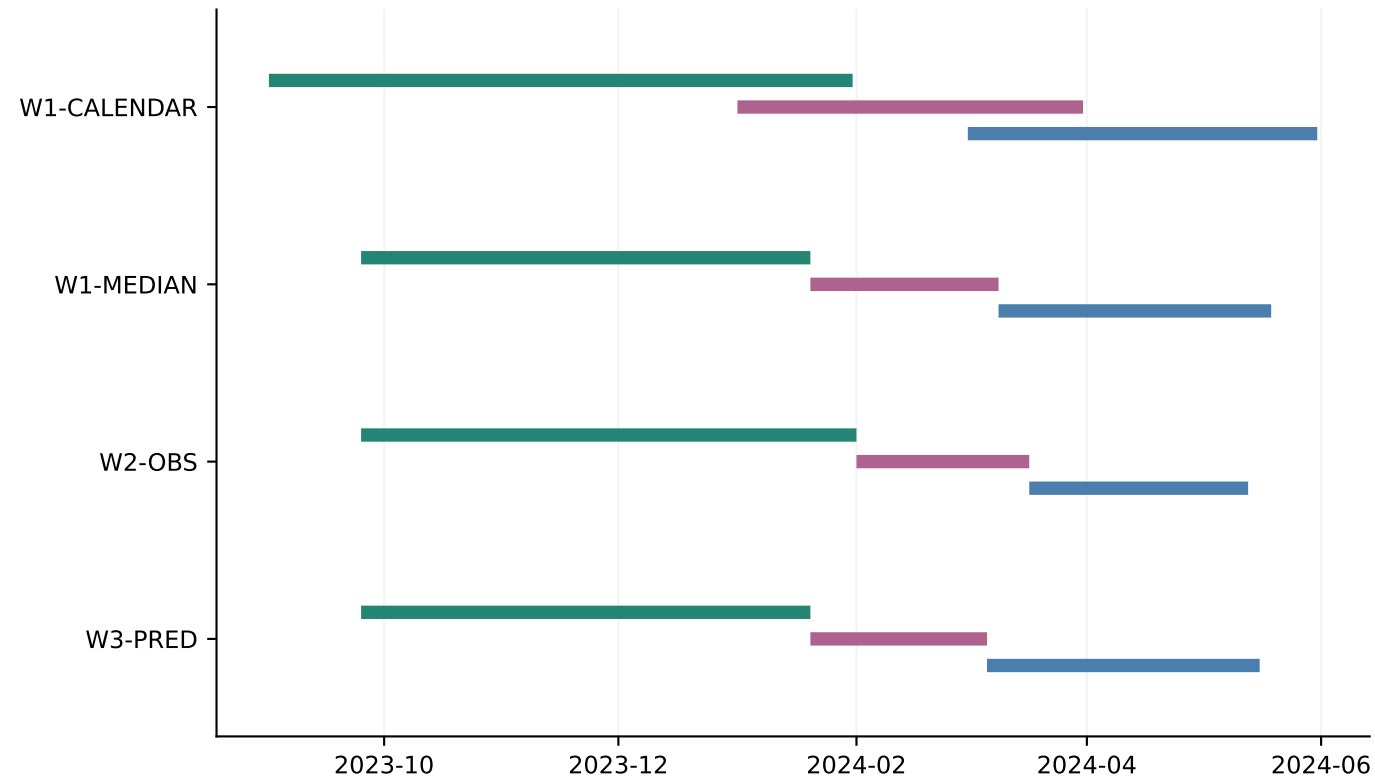


## P3-M1 | days from harvest Jan 01



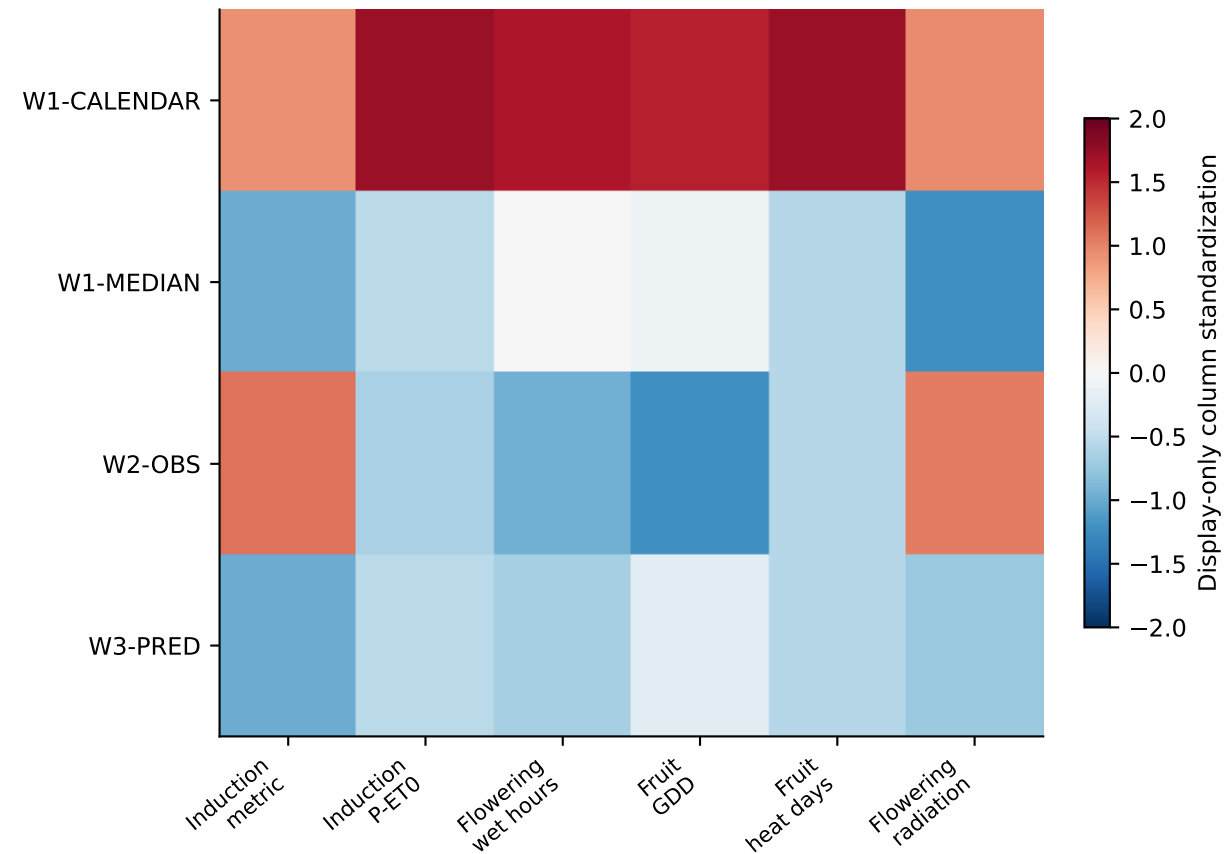
● Bannei    ▲ Hongming    ■ Luhong

**Bannei 2024: held-out window boundaries**



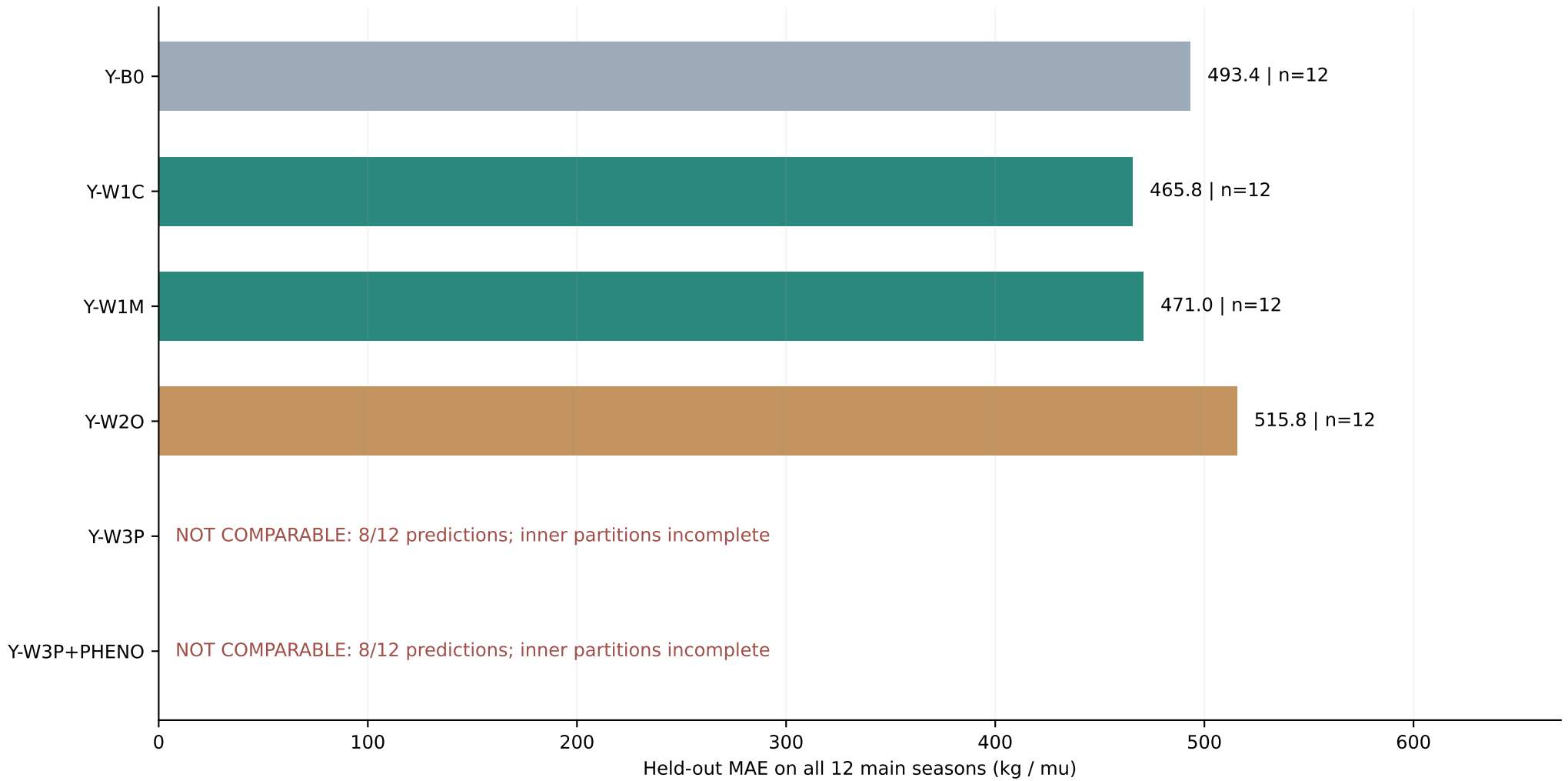
induction    flowering    fruit development

**Feature differences in the same example**



2.0  
1.5  
1.0  
0.5  
0.0  
-0.5  
-1.0  
-1.5  
-2.0  
Display-only column standardization

# Normal yield: no demonstrated advantage of observed dynamic windows



# P2-only fixed-model temperature sensitivity

## Conditional on the observed panicle date, not a complete-chain forecast

